AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 34 · STUDENT LESSON

Capstone Dataset and Baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Weeks 8,10 and 24: group identities and fit the baseline on training only.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Define a feasible capstone question and acceptance criteria. Try the recall before opening notes.

Your learning target

I can define a feasible capstone question and acceptance criteria.

I can create a data and split plan before modelling.

I can specify **evidence**, privacy and a fallback.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A responsible capstone begins by freezing the problem, intended use, dataset documentation, grouped split, and **baseline evidence**.



Your target

LEARNING TARGET A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and **baseline** procedure. When may final labels be consulted?

Builder name and experiment date:

Readiness for the connected explanation

Which evaluation choices should precede model selection?

Readiness for Capstone models and baselines

What must be fixed before the capstone comparison?

My readiness explanation

This week's end goal

By the end, I can explain capstone dataset and **baseline** using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Capstone Dataset and Baseline case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use capstone dataset and **baseline** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

The capstone is a classroom investigation, not a product launch. Choose the supplied fictional clear/blocked parcel task or an equally bounded teacher-approved synthetic task.

Write a question that can be answered by comparing a simple model against a **baseline**. Do not use private student records or real safety decisions.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines capstone data and **requirement**s.

Apply the connected idea: Capstone models and baselines

Build the **baseline** before the candidate. For classification, use the most common **training** class, with a declared tie policy.

Choose one simple candidate using techniques already learned, such as k-nearest neighbours with **training**-only scaling and a fixed k. Record every development attempt; complexity earns no automatic credit.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines capstone models and **baseline**s.

STUDENT LESSON · 3 OF 18

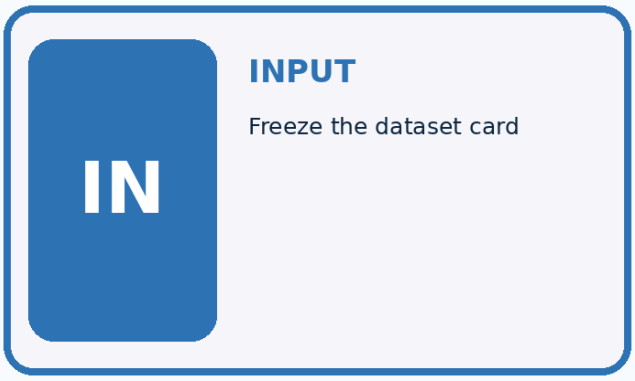
See how it works

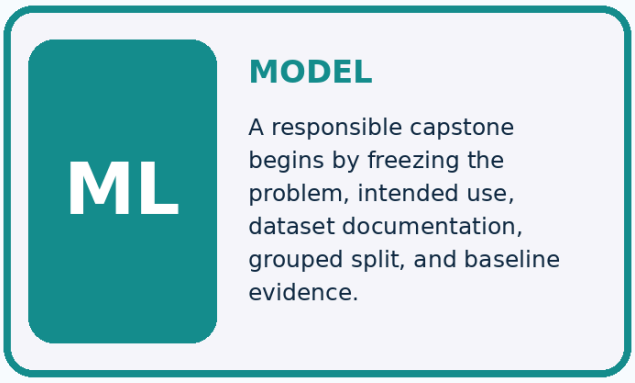
> Follow the mechanism for Capstone Dataset and Baseline. Connect each labelled input, deciding step and result.

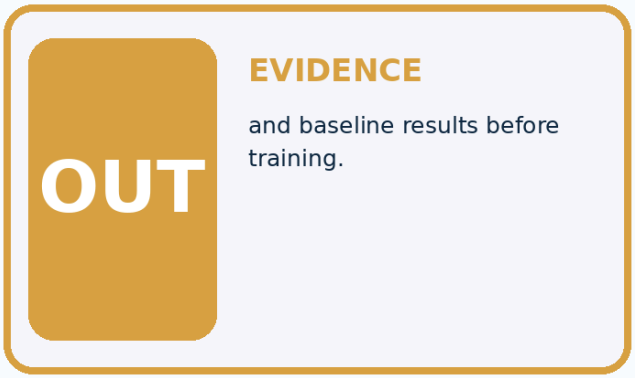
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION **Freeze** the dataset card, split, and **baseline result**s before **training**.

MODEL IDEA A responsible capstone begins by freezing the problem, intended use, dataset documentation, grouped split, and **baseline evidence**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets. Test set remains sealed; validation supports model choices.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Example proposal: can a small classifier improve validation accuracy over the **training**-majority **baseline** on fictional parcel measurements?

Before modelling, assign 40 **training**, 20 validation and 12 final-test records, with related parcel groups kept together.

Define blocked as positive, keep a fixed label rule, and reserve test labels with the teacher until Week 35.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Your teacher can prepare the 72-record fictional grid described in the teacher guide, or approve another documented synthetic dataset. Each supplied record has an ID, group ID, width and height in cm.

Training and validation labels are released for development; final-test labels stay with the teacher until the model is frozen.

The numerical results in later worked examples are illustrative and are not expected scores for your project.

Apply the connected idea: Capstone models and baselines

Illustrative **training** has 30 clear and 10 blocked cases, so the **baseline** always predicts clear. On 20 validation cases with 14 clear and 6 blocked, its matrix is[[14, 0],[6, 0]], accuracy 70 percent.

Candidate matrix[[12, 2],[2, 4]] gives accuracy 80 percent, precision 4/6 and recall 4/6. These are example results, not the student team results.

```
# A check of the illustrative validation matrix.
# Replace with your own saved counts; do not invent results.
tn, fp, fn, tp = 12, 2, 2, 4
n = tn + fp + fn + tp
print("accuracy", (tn + tp) / n)
print("precision", tp / (tp + fp))
```

```
print("recall", tp / (tp + fn))
# Compare with the training-majority baseline: 14 / 20.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

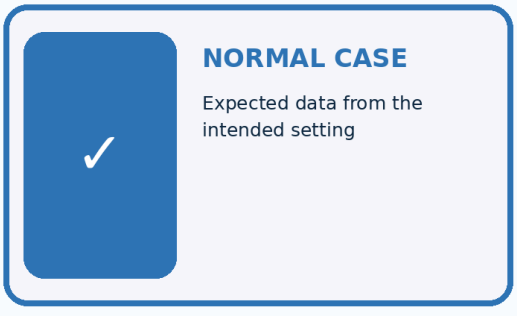
Compare the cases

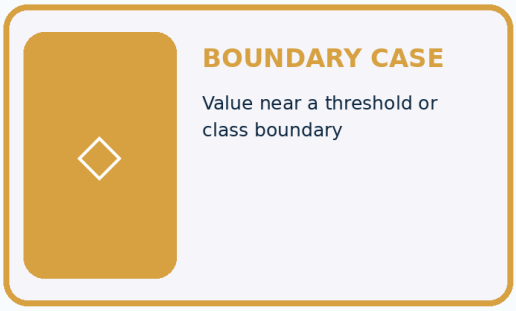
> Compare the supplied cases for Capstone Dataset and Baseline. Identify the detail that changes the conclusion.

EVALUATION DESIGN

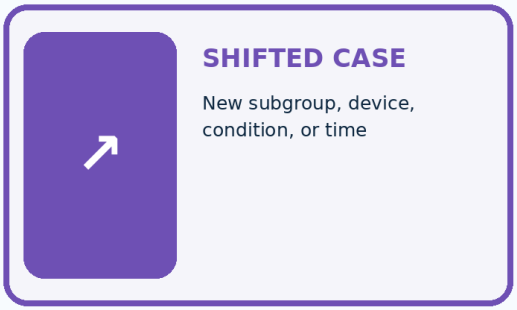
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: A derived crop of plantG is added to **training**.

Predicted result

Prediction: The final group separation is violated.

Why the result follows

Reason: The crop retains the same source plant identity even with a new file name.

Words that unlock the idea

> Use the vocabulary of Capstone Dataset and Baseline to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
freeze	lock a reviewed version for comparison	
acceptance test	observable required behavior	
baseline result	simple method's measured performance	
evidence bundle	linked artifacts supporting a claim	

MISCONCEPTION TO REPAIR Capstone work should begin by **training** the most advanced available model.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Requirement	A concrete condition the project must meet
Acceptance criterion	Evidence that decides whether a requirement is met
Data dictionary	Definitions and units for dataset fields
Milestone	A review point with a stated deliverable

Terms for Capstone models and baselines

Term	Meaning
Baseline	A simple reference method fitted or chosen from training
Prediction ledger	A per-record table of reference and predicted outputs
Frozen artifact	A saved model and procedure that will not change during testing
Decision record	A written reason for a development choice

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Capstone Dataset and Baseline, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Four-step worked evidence for Week 34: Capstone Dataset and Baseline

STEP	EVIDENCE AND REASONING
1	Freeze dataset card v3 and grouped split s2 with no plant identity crossing sets.
2	Test set remains sealed; validation supports model choices.
3	Validation has 12 healthy and 8 spotted leaves. The always-healthy baseline scores 12/20=60% accuracy and 0/8=0% spotted recall. The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.
4	Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Step 1

STEP 1 **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets.

Step 2

STEP 2 Test set remains sealed; validation supports model choices.

Step 3

STEP 3 Validation has 12 healthy and 8 spotted leaves. The always-healthy **baseline** scores 12/20=**60%** accuracy and 0/8=0% spotted recall. The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.

Step 4

STEP 4 Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Capstone data and requirements at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Week 33	Task data and split plan	Teacher checks feasibility
Week 34	Baseline and candidate	Validation decision recorded
Weeks 35 and 36	Frozen test and defense	Evidence plus limitations

The data dictionary lists record ID, group ID, width in cm, height in cm and reference class. IDs track provenance and splitting; they are not automatic predictive inputs. Check uniqueness, units, missing values, class coverage and group separation. Fit imputation or scaling only on training.

Explain why record ID, group ID and measured width have different roles. Explain the steps or evidence behind your response.

Apply the connected idea: Capstone models and baselines

Capstone models and baselines at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Training	Baseline always clear	30 clear and 10 blocked
Validation baseline	14 correct of 20	6 blocked cases missed
Validation candidate	16 correct of 20	2 false alarms 2 misses

The candidate gains two correct cases while introducing two false alarms and reducing misses from six to two. Whether that tradeoff is acceptable depends on the predeclared classroom criteria. Do not choose a threshold or metric after the final test. If using a regression alternative, compare with a training-derived constant and report MAE in target units.

Compute the worked candidate accuracy, precision and recall. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone Dataset and Baseline

Plan the plant capstone around independent groups and a **baseline** before collecting more rows.

The facts we will use

Supplied facts

Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H. Target is visible-spot status, with review for hidden leaves.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/34

Capstone Dataset and Baseline

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define target

Write visible-spot and review rules before annotation.



2. Split by groups

Keep all 18 A-F views in development and all 6 G/H views for final evaluation.



3. Plan baseline

Fit majority label only from development training; record metrics and error types.

CHANGE ONE THING

A derived crop of plantG is added to training.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define target	Write visible-spot and review rules before annotation.
2. Split by groups	Keep all 18 A–F views in development and all 6 G/H views for final evaluation.
3. Plan baseline	Fit majority label only from development training; record metrics and error types.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A well-planned split does not create sufficient sample size; two held-out plants support only

a very narrow claim.

Keep the stated limit

Read the labelled picture: What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: A derived crop of plantG is added to **training**. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Capstone Dataset and Baseline claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Capstone work should begin by training the most advanced available model. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Capstone work should begin by **training** the most advanced available model.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

Submit a one-page proposal, **data dictionary**, split-membership record and risk/limitations note. The teacher approves feasibility and synthetic-data scope before modelling.

If there are too few valid records for an independent test, use a clearly labelled demonstration and explicitly limit the claim rather than inventing **evidence**.

What should the team do if it cannot obtain a valid untouched final test?

Apply the connected idea: Capstone models and baselines

Check the claim against the facts

The deliverable is a runnable or fully traceable **baseline**/candidate comparison, a validation table and a signed-off **freeze** record. If the **baseline** wins, retain it and explain why.

If a pipeline fails, record the failure and fix it on development data before requesting final testing.

List what must be saved before the teacher releases final-test labels.

STUDENT LESSON · 8 OF 18

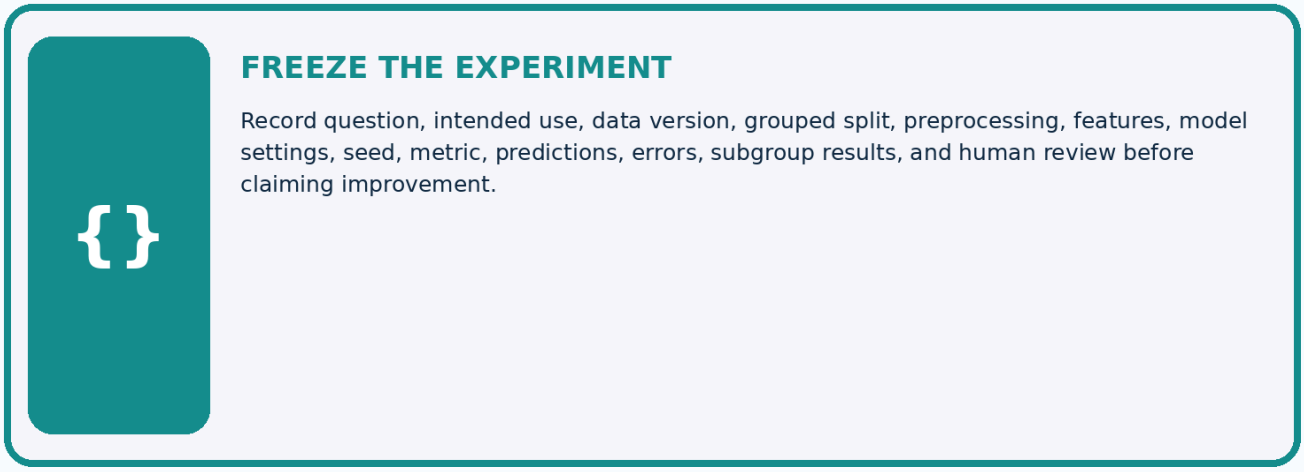
Practise together

Practise Capstone Dataset and Baseline with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Acceptance criteria describe the **process** even if the model loses: no split leakage; **prediction**s for every planned case or documented failures; the same metric for **baseline** and candidate; precision/recall with denominators; a model card with limits.

Improvement is a research outcome, not a grade **requirement**.

Apply the connected idea: Capstone models and baselines

Save the data version, feature order, pre**process**ing values, model settings, tie policy and **prediction**s. A **prediction ledger** includes row ID, split, reference, **baseline output**, candidate **output** and error type.

Test labels remain held by the teacher. A peer should be able to reproduce one row by hand.

Practise with less help: Define fields

Use workbook W2 for the connected example “Capstone data and **requirement**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone data and **requirement**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Capstone models and baselines

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A final answer is enough

Explain the picture

What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Capstone Dataset and Baseline. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Capstone Gate 1: peer-audit dataset card, leakage, split, baseline calculations, and acceptance tests; repair before build.

1. Write the hypothesis and expected **evidence** before running.

2. Capstone Gate 1: peer-audit dataset card, leakage, split, **baseline** calculations, and **acceptance test**s; repair before build.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

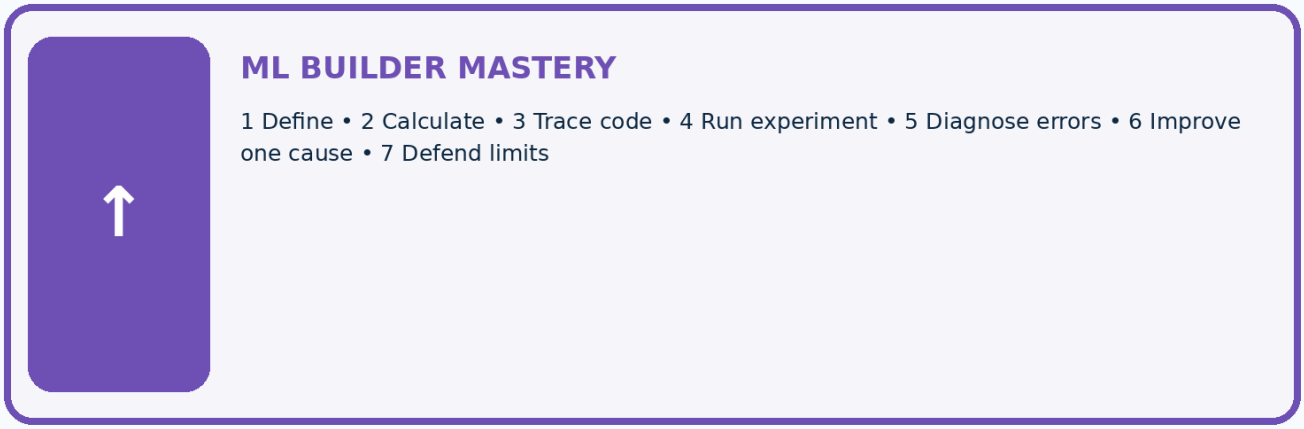
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Capstone Dataset and Baseline. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **freeze**, **acceptance test**, **baseline result**, **evidence bundle**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Capstone work should begin by **training** the most advanced available model.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Publish the frozen Plant Classifier design-and-data bundle with version IDs and approval record.

THIS WEEK'S CHECKPOINT Publish the frozen Plant Classifier design-and-data bundle with version IDs and approval record.

Which **requirement** or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

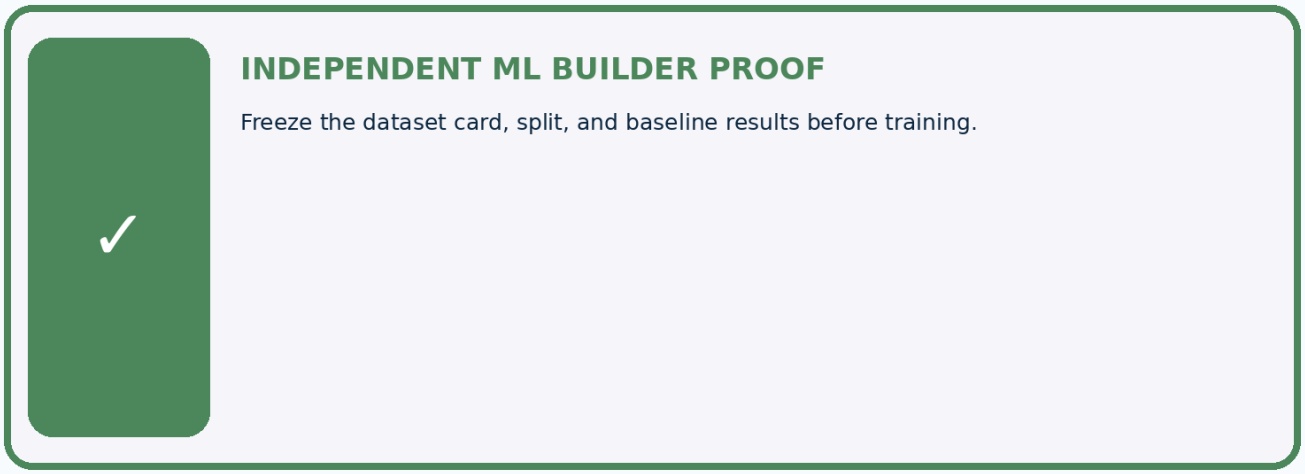
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF **Freeze** the dataset card, split, and **baseline result**s before **training**.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Capstone Train, Analyze, Improve.

Study a complete case

> Revisit the complete Capstone Dataset and Baseline case for supported practice. Repeated examples do not become fresh assessments.

Week 34 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Capstone has60 independent leaves, several photos per leaf, healthy/spotted labels.

The question

Plan grouping and **baseline**.

Worked explanation

Keep all views together in train/validation/test partitions. Define labels/use, fit a majority **baseline** on **training** labels and evaluate declared metrics.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Begin the capstone with independent groups

Evidence or stage	Value or result
Dataset	Sixty leaves, multiple views each
Group	Keep views of one leaf together
Partition	Separate train, validation and test
Baseline	Fit majority choice on training labels

Capstone has60 independent leaves, several photos per leaf, healthy/spotted labels.

Plan grouping and **baseline**.

Keep all views together in train/validation/test partitions. Define labels/use, fit a majority **baseline** on **training** labels and evaluate declared metrics.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

A proposed 80-row dataset has 40 related pairs, and the team splits individual rows randomly then uses pair ID as a feature. Revise its plan.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 34 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Your capstone dataset includes duplicate photographs in both **training** and validation. What should be repaired before measuring the **baseline**?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Publish the frozen Plant Classifier design-and-data bundle with version IDs and approval record.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and **baseline** procedure. When may final labels be consulted?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Capstone Dataset and Baseline, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Plan the plant capstone around independent groups and a **baseline** before collecting more rows.

By the end, I can	How I will show it
Define a feasible capstone question and acceptance criteria	Explain the deciding step, show a trace or test, and state one limit.
Create a data and split plan before modelling	Explain the deciding step, show a trace or test, and state one limit.
Specify evidence, privacy and a fallback	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Capstone Dataset and Baseline without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Define a feasible capstone question and acceptance criteria

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plan the plant capstone around independent groups and a **baseline** before collecting more rows.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Fit majority label only from development **training**; record metrics and error types.

1. Define target

2. Split by groups

3. Plan **baseline**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Plan **baseline** Fit majority label only from development **training**; record metrics and error types. Read the picture in order.

Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A derived crop of plantG is added to **training**. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The final group separation is violated.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The final group separation is violated. The crop retains the same source plant identity even with a new file name.

Repair a missing building block

Review Model Cards and Human Review (ai-4/33). Goal: Write a model card whose claims match **evidence**. Short practice: Write a bounded replacement for the publicity sentence.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Subgroup Performance (ai-4/31). Goal: Calculate subgroup results with sample sizes. Short practice: Why should the report retain subgroup rates alongside the total?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What comparison is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

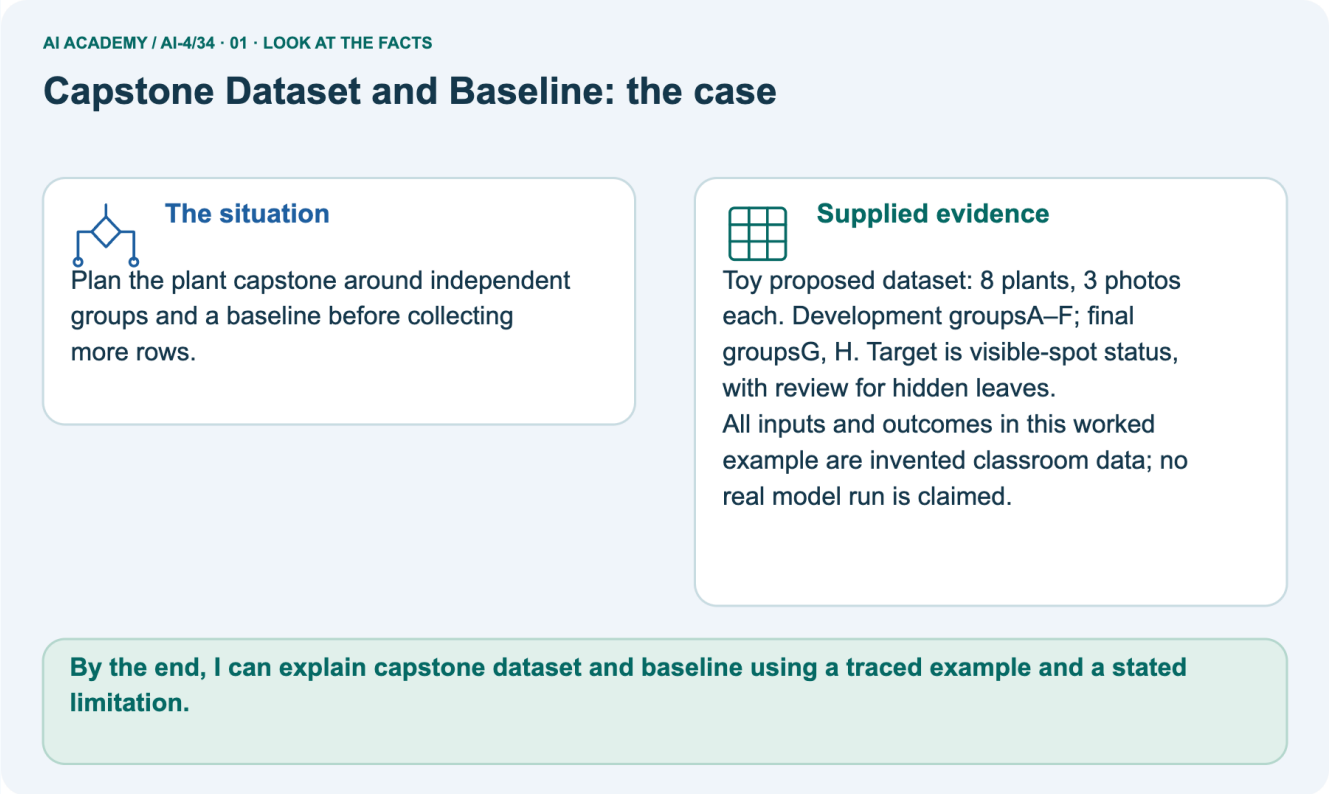
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Capstone Dataset and Baseline graphic using its labels, deciding evidence and relevant limitation.

01. Capstone Dataset and Baseline: the case



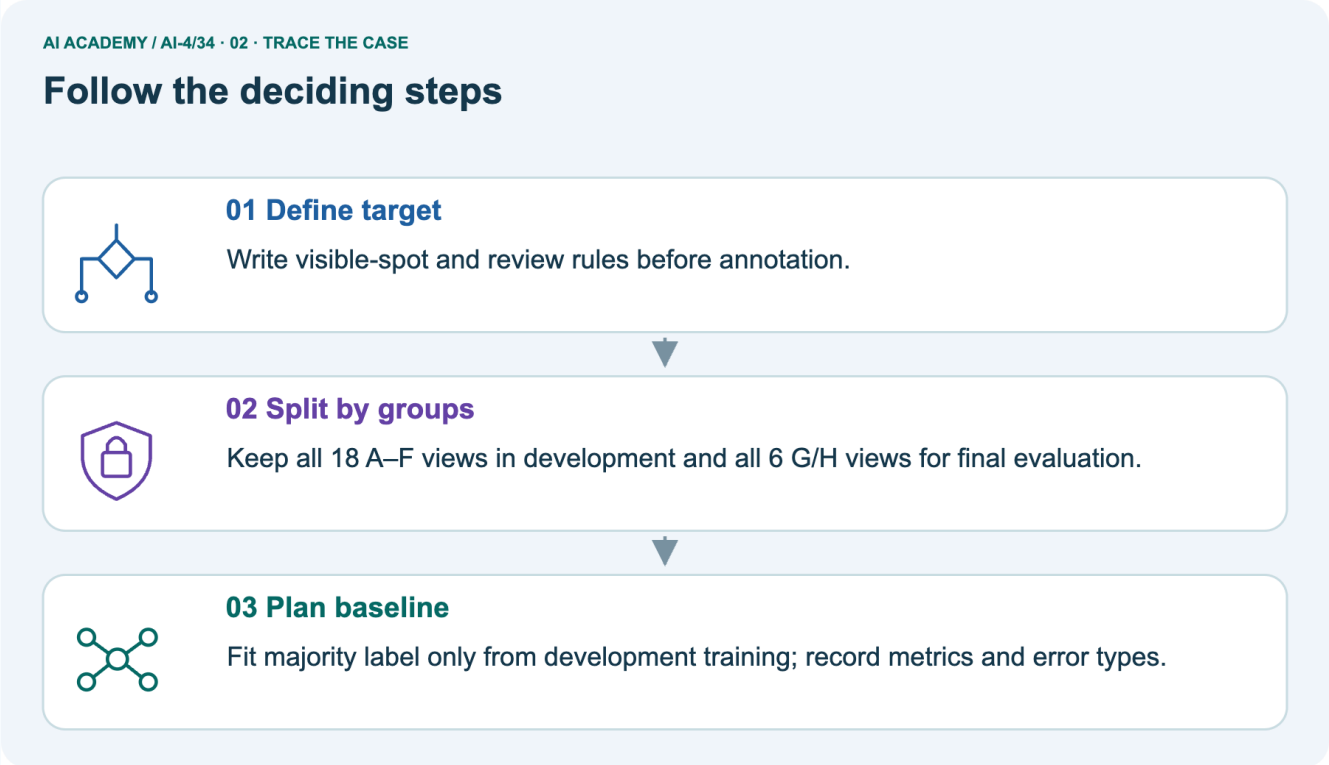
Follow the labelled picture with the matching explanation. Open the vector diagram

Plan the plant capstone around independent groups and a **baseline** before collecting more rows. Supplied facts: Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H.

Target is visible-spot status, with review for hidden leaves. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Define target: Write visible-spot and review rules before annotation. → Split by groups: Keep all 18 A–F views in development and all 6 G/H views for final evaluation. → Plan **baseline**: Fit majority label only from development **training**; record metrics and error types.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Four-step worked evidence for Week 34: Capstone Dataset and Baseline

AI ACADEMY / AI-4/34 · 03 · INSPECT THE LESSON MODEL

Four-step worked evidence for Week 34: Capstone Dataset and Baseline

Step	Evidence and reasoning
1	Freeze dataset card v3 and grouped split s2 with no plant identity crossing sets.
2	Test set remains sealed; validation supports model choices.
3	Validation has 12 healthy and 8 spotted leaves. The always-healthy baseline scores 12/20=60% accuracy and 0/8=0% spotted recall. The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.
4	Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Follow the labelled picture with the matching explanation. Open the vector diagram

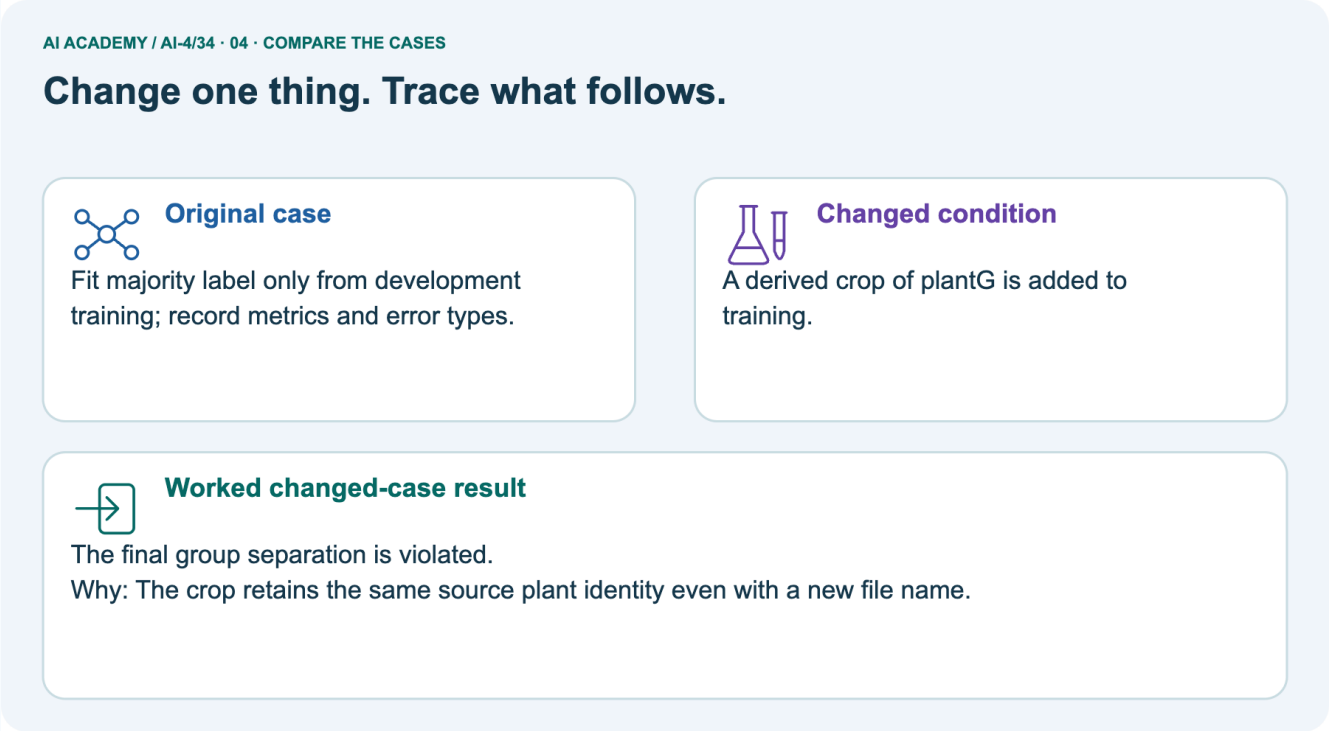
Four-step worked **evidence** for Week 34: Capstone Dataset and **Baseline**. 1 | **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets.. 2 | Test set remains sealed; validation supports model choices.. 3 | Validation has 12 healthy and 8 spotted leaves.

The always-healthy **baseline** scores 12/20=**60%** accuracy and 0/8=0% spotted recall.

The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.. 4 | Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



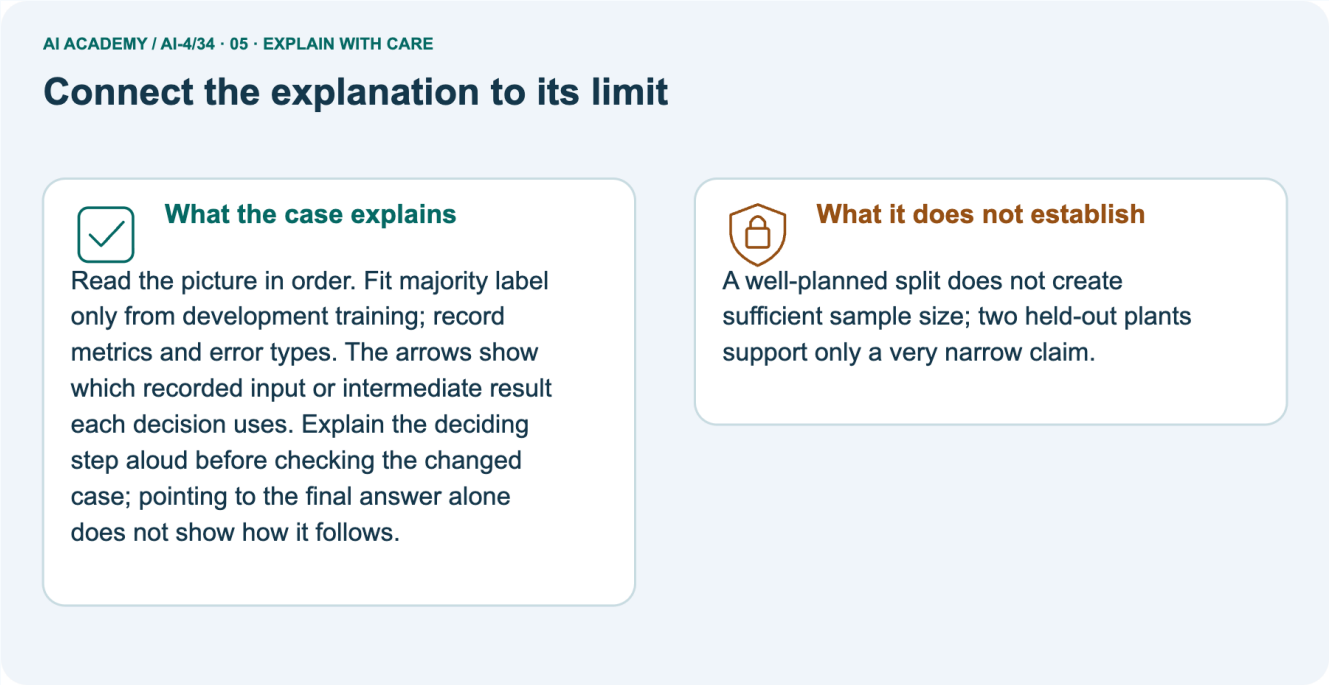
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Fit majority label only from development **training**; record metrics and error types. Changed condition: A derived crop of plantG is added to **training**. Expected result: The final group separation is violated.

Reason: The crop retains the same source plant identity even with a new file name.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

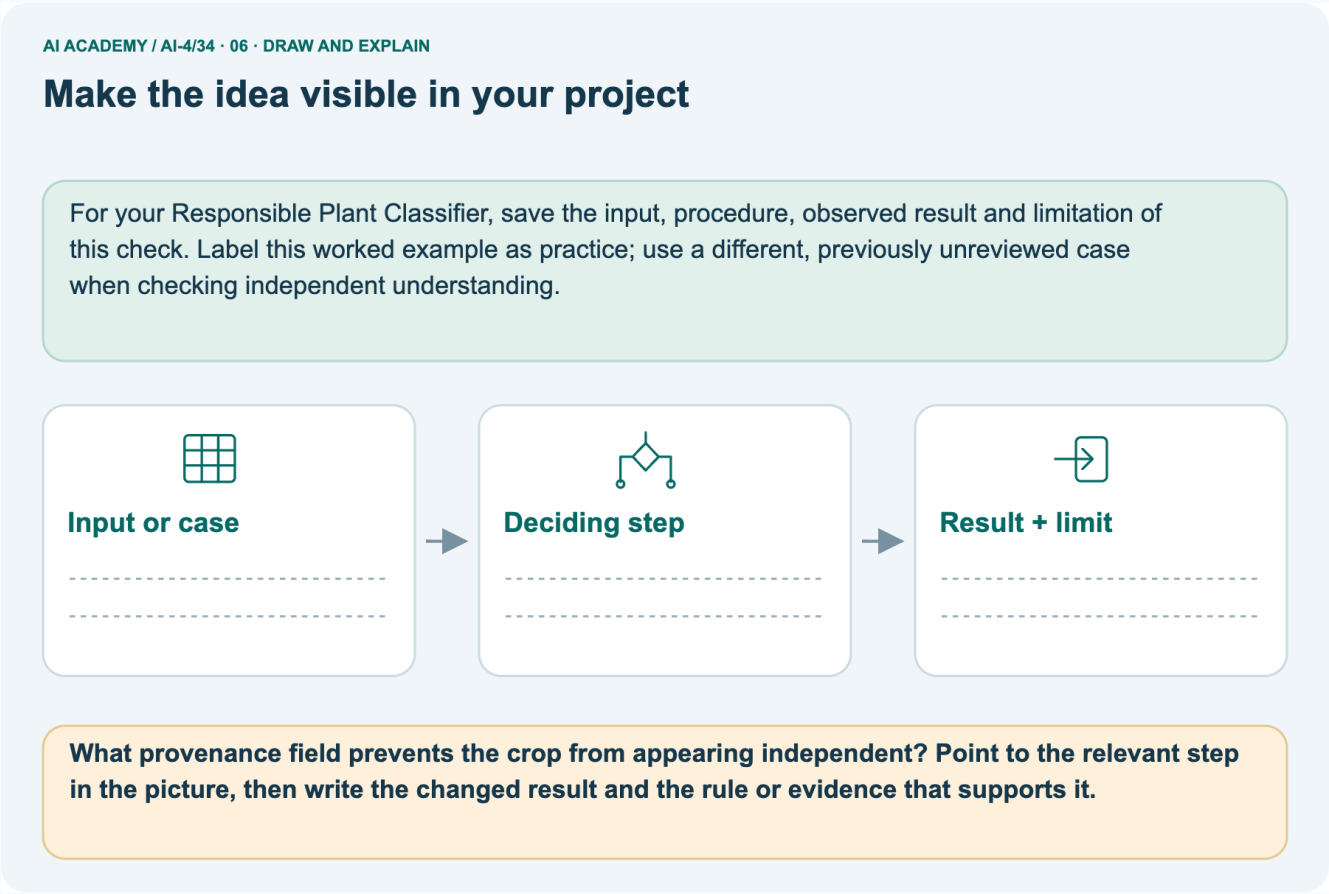
Read the picture in order. Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A well-planned split does not create sufficient sample size; two held-out plants support only a very narrow claim.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.